

Efficacy and Safety of Semaglutide versus Liraglutide for Weight Management in Adults with Overweight or Obesity: The STEP-8 Randomized Clinical Trial

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ABSTRACT

BACKGROUND Glucagon-like peptide-1 (GLP-1) analogues are effective for weight management. However, direct comparisons between once-weekly subcutaneous Semaglutide 2.4 mg and once-daily subcutaneous Liraglutide 3.0 mg are limited. **METHODS** We conducted a randomized, open-label, 68-week trial at 19 sites involving adults with a body-mass index (BMI) of 30 or greater, or 27 or greater with one or more weight-related comorbidities, and without diabetes. Participants were randomly assigned (3:1:3:1) to receive subcutaneous Semaglutide 2.4 mg once weekly, matching placebo, subcutaneous Liraglutide 3.0 mg once daily, or matching placebo. All received diet and physical activity counseling. **RESULTS** The mean weight change from baseline to week 68 was -15.8% with Semaglutide vs -6.4% with Liraglutide (difference, -9.4 percentage points; 95% CI, -12.0 to -6.8; $P < 0.001$). Participants who received Semaglutide had greater improvements in cardiometabolic risk factors than those who received Liraglutide. Gastrointestinal adverse events were common with both treatments (84.1% Semaglutide vs 82.7% Liraglutide). **CONCLUSIONS** Among adults with overweight or obesity without diabetes, once-weekly subcutaneous Semaglutide resulted in significantly greater weight loss than once-daily subcutaneous Liraglutide.

INTRODUCTION

Obesity is a complex, chronic disease associated with numerous complications, including Type 2 Diabetes, cardiovascular disease, obstructive sleep apnea, and certain cancers. Despite public health efforts, the prevalence continues to rise globally. Lifestyle interventions, including diet and exercise, remain the cornerstone of weight management but frequently result in insufficient long-term weight loss due to metabolic adaptation and hormonal upregulation of appetite.

Pharmacotherapy is recommended as an adjunct to lifestyle intervention for individuals with a BMI of 30 or higher, or 27 or higher with comorbidities. Glucagon-like peptide-1 (GLP-1) receptor agonists mimic the effects of endogenous GLP-1, a gut hormone that regulates appetite and food intake. Liraglutide 3.0 mg, a once-daily GLP-1 analogue, was approved for weight management in 2014 based on the SCALE program. Semaglutide 2.4 mg, a once-weekly GLP-1 analogue, has recently demonstrated superior efficacy in the STEP (Semaglutide Treatment Effect in People with obesity) clinical trial program. This head-to-head trial (STEP-8) was designed to compare their efficacy and safety directly.

METHODS

TRIAL DESIGN AND OVERSIGHT This was a 68-week, randomized, open-label, active-controlled, parallel-group, phase 3b trial conducted at 19 sites in the United States. The trial was conducted in strict accordance with the Declaration of Helsinki and Good Clinical Practice

guidelines. The protocol was approved by independent ethics committees at each site.

PARTICIPANTS Eligible participants were adults (≥ 18 years) with a stable body weight and a BMI of ≥ 30 , or ≥ 27 with at least one weight-related comorbidity (e.g., hypertension, dyslipidemia, obstructive sleep apnea, or cardiovascular disease). Patients with Type 1 or Type 2 Diabetes were excluded to isolate the weight-loss effect from glycemic control. Key exclusion criteria included a history of pancreatitis, major depressive disorder, or use of other weight-loss medications within 90 days of screening.

INTERVENTIONS AND PROCEDURES Participants were randomized to Semaglutide 2.4 mg once weekly or Liraglutide 3.0 mg once daily. Doses were escalated over 16 weeks (Semaglutide) or 4 weeks (Liraglutide) to minimize gastrointestinal side effects. Both groups received standard lifestyle counseling, including a deficit of 500 kcal/day and 150 minutes of physical activity per week.

STATISTICAL ANALYSIS The primary endpoint was the percentage change in body weight from baseline to week 68. The confirmatory secondary endpoint was achievement of weight loss of 10% or more, 15% or more, and 20% or more. Efficacy analyses were based on the treatment policy estimand using a mixed model for repeated measures. A sample size of 338 provided 90% power to detect a difference of 5 percentage points between groups.

RESULTS

WEIGHT LOSS OUTCOMES A total of 338 participants underwent randomization (126 to Semaglutide, 127 to Liraglutide, and 85 to placebo groups combined). Baseline characteristics were similar across groups; the mean age was 49 years, mean body weight was 104.5 kg, and mean BMI was 37.5. 78.4% were female.

The mean change in body weight from baseline to week 68 was -15.8% (Standard Deviation [SD] 6.0) in the Semaglutide group compared with -6.4% (SD 5.0) in the Liraglutide group. The estimated treatment difference was -9.4 percentage points (95% CI, -12.0 to -6.8; $P < 0.001$).

Treatment with Semaglutide lead to significantly higher rates of categorical weight loss: - Weight loss $\geq 10\%$: 70.9% (Semaglutide) vs 25.6% (Liraglutide) - Weight loss $\geq 15\%$: 55.6% (Semaglutide) vs 12.0% (Liraglutide) - Weight loss $\geq 20\%$: 38.5% (Semaglutide) vs 6.0% (Liraglutide)

CARDIOMETABOLIC RISK FACTORS Improvements in waist circumference, systolic and diastolic blood pressure, C-reactive protein, and lipid levels were greater with Semaglutide than with Liraglutide. For example, the reduction in waist circumference was 13.5 cm with Semaglutide vs 6.8 cm with Liraglutide ($P < 0.001$). Systolic blood pressure decreased by 5.5 mm Hg with Semaglutide compared to 3.2 mm Hg with Liraglutide.

SAFETY AND ADVERSE EVENTS

Adverse events were reported by 95.2% of participants in the Semaglutide group and 96.1% in the Liraglutide group. The most common adverse events were gastrointestinal disorders: nausea (60.3% vs 59.1%), vomiting (35.7% vs 31.5%), diarrhea (30.2% vs 24.4%), and constipation (23.8% vs 31.5%). Most events were mild to moderate in severity and transient, occurring primarily during the dose-escalation phase.

Serious adverse events occurred in 7.9% of participants in the Semaglutide group and 11.0% in the Liraglutide group. Treatment discontinuation due to adverse events occurred in 13.5% of participants with Semaglutide and 27.6% with Liraglutide, suggesting better overall tolerability of the weekly regimen despite similar GI side effect rates.

DISCUSSION

In this head-to-head trial, once-weekly Semaglutide 2.4 mg was superior to once-daily Liraglutide 3.0 mg for weight reduction in adults with overweight or obesity without diabetes. The magnitude of weight loss observed with Semaglutide (approx. 16%) approaches that seen with bariatric surgery (typically 20-30%).

The mechanistic basis for the greater efficacy of Semaglutide may relate to its unique pharmacokinetic profile derived from albumin binding, allowing for sustained GLP-1 receptor activation, and potentially greater penetration into specific brain regions regulating appetite and satiety compared to Liraglutide.

While both drugs are effective, the magnitude of weight loss with Semaglutide sets a new benchmark for pharmacotherapy. Tirzepatide, a dual GIP/GLP-1 agonist, has shown even greater efficacy in recent trials (SURMOUNT-1), but was not included in this comparison.

LIMITATIONS The trial was open-label, which may introduce bias. The duration was 68 weeks, and longer-term data are needed to assess durability of weight loss and long-term safety. The majority of participants were female and White, which may limit generalizability to other populations.

REFERENCES

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